SCHOOL OF COMPUTING TECHNOLOGY AND APPLIED SCIENCES

CCS2131 DATA STRUCTURES AND ALGORITHMS

SEMESTER FINAL EXAMINATION

FRIDAY, 5 JUNE 2026

09:00 – 12:00 HOURS

TIME ALLOWED : WRITING — THREE HOURS : READING —5 MINUTES

INSTRUCTIONS:

1. Section A: this question is compulsory and must be attempted.

2. Sections B: Answer THREE (3) questions from this section.

3, This examination paper carries a total of 100 marks.

4. Candidates must not turn this page until the invigilator tells them to do so.

Section A

Question 1

a) A hospital emergency department uses a triage priority score to decide which patients should be attended to first. Each patient is assigned a score based on the seriousness of their condition, waiting time, and risk level. A higher score means the patient needs more urgent attention.

During one shift, the triage desk records the following priority scores for eight patients:

[58,92,31,77,46,85,19,64]

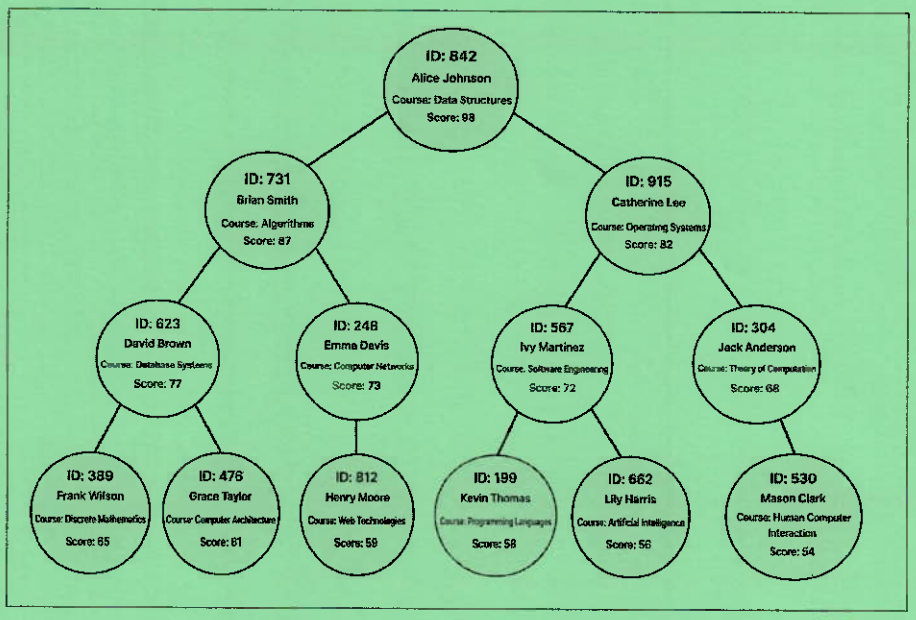
The hospital wants to arrange these scores using a maximum heap tree so that the patient with the highest priority score is always attended to first.

Using the priority scores given above, construct a maximum heap tree and show the final heap arrangement after all values have been inserted. Show all your workings. **(15 Marks)**

b) Apply the following tree traversal algorithms to the student course record tree below using the ID:

1. Pre-order traversal
2. In-order traversal

For each traversal, list and illustrate how the student records will be visited:



(10 Marks)

c) A retail supermarket records the total daily sales, in Kwacha, from 10 different branches. The sales figures were entered into the system in the order they were received:

[12500, 8400, 19300, 6700, 15800,9200, 21100, 13400, 7600, 17600]

The head office wants to arrange the branch sales in ascending order to identify the branches with the lowest and highest daily sales.

Using the merge sort algorithm, sort the sales figures in ascending order. Show all your workings. **(10 Marks)**

d) Explain the concept of a self-organizing list in data structures and describe how it improves search performance. **(5 Marks)**

[Total 40 Marks]

Section B

Question 2

a) A delivery company uses the following Python code to calculate the total weight of packages loaded into a van:

```
1.  Weights = [12, 8, 15, 6, 10, 20, 5]
2.  total_weight = 0

3.  for weight in weights:
4.          total_weight = total_weight + weight

5.  print(total_weight)
```

Calculate the time complexity of the code above using Big-O notation. **(10 Marks)**

b) Suppose each letter of the alphabet is assigned a value from 1 to 26, as follows:

a=1,b=2,c=3,…,z=26

A computer system stores characters in a hash table using the modulus operation to calculate the hash index:

Hash Index=Character Value mod n

Using the string:

"Secure Your Data "

Assume that:

1. Spaces, digits, and quotation marks are ignored.
2. The input is not case-sensitive, meaning S and s are treated the same.

3. The hash table has 13 slots, indexed from 0 to 12.
4. If a collision occurs, use linear probing to find the next available slot.

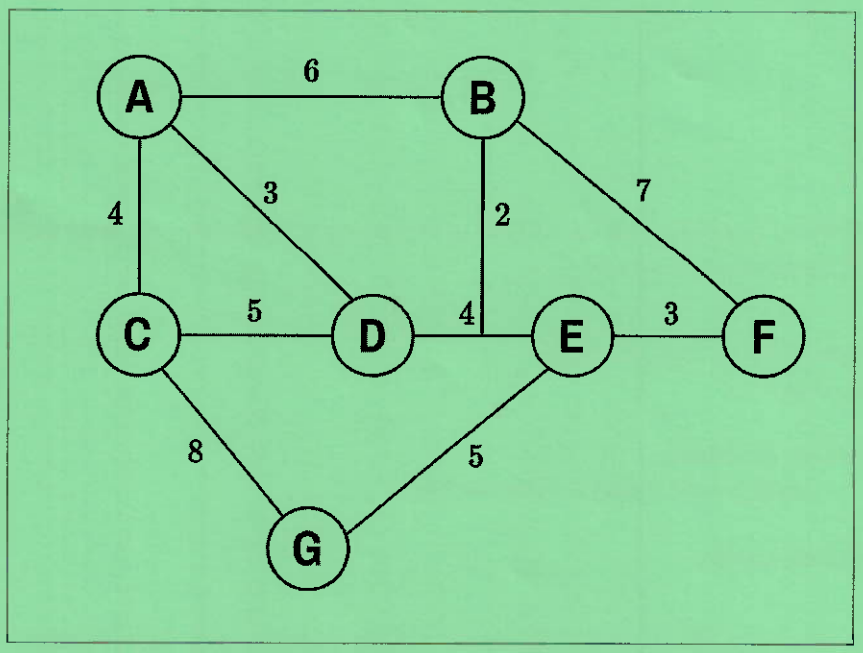
Insert the characters from the string into the hash table using their assigned alphabet values. Show the final contents of the hash table after all insertions.

(10 Marks)

[Total 20 Marks]

Question 3

a) A transport company wants to find the shortest route from its main depot to a customer delivery point. The road network is represented by the weighted graph below. Each node represents a location, and each edge weight represents the distance in kilometres.



Using Dijkstra's algorithm, find the shortest path from node A to node F.

In your answer, show:

1. The order in which the nodes are selected.
2. The shortest distance updates.
3. The final shortest path from A to F.
4. The total shortest distance. **(10 Marks)**

b) Write a Python program that uses **recursion** to count down from a positive number entered by the user to 1.

Your program should:

1. Define a recursive function called countdown(n).

2. Use a base case to stop the recursion.
3. Allow the function to call itself.
4. Display each number during the countdown.

Example:
If the user enters 5, the program should display: 5,4,3,2,1

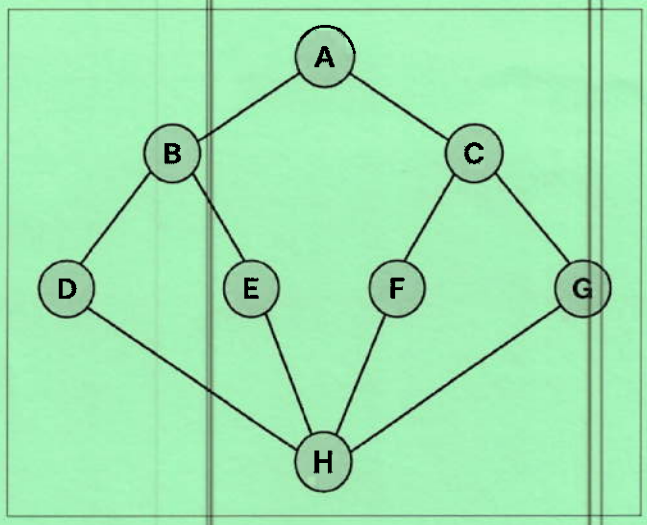
(10 Marks)

[Total 20 Marks]

Question 4

a) The following graph represents connected locations in a small computer network. Each node represents a computer, and each edge represents a direct connection between two computers.

Starting from node A, perform a Depth First Search (DFS) traversal of the graph using a stack. Show all your workings. **(10 Marks)**



b) A Software developer has written two different algorithms to search for a customer record in a company database. Both algorithms give the correct result, but one becomes very slow when the number of customer records Increases. Explain why it is important to analyse algorithms before choosing one for implementation. **(5 Marks)**

c) Identify and briefly explain five basic operations that can be performed on a queue data structure. **(5 Marks)**

[Total 20 Marks]

Question 5

a) A music playlist application stores songs using a doubly linked list. Each song record has a link to the next song and a link to the previous song, allowing the user to move forward and backward through the playlist. List five operations that can be performed on the doubly linked list used in the playlist application. **(6 Marks)**

b) A web browser uses a stack to keep track of pages visited by a user. The most recently visited page is placed on top of the stack, and when the user clicks the Back button, that page is removed first. With the aid of a practical example, demonstrate three common operations that can be performed on a stack. **(7 Marks)**

c) A company uses different data structures to organise information. A binary tree is used to represent an employee reporting structure, where each manager can supervise at most two employees. A graph is used to represent a transport network, where locations are connected by several possible routes. With the aid of examples, distinguish between binary tree and graph data structures. **(7 Marks)**

[Total 20 Marks]

END OF EXAMINATION